Fiscal Year:	FY 2020	**Task Last Updated:**	FY 04/09/2020
PI Name:	German, Randall M. Ph.D.		
Project Title:	Multi-Scale Modeling and Experimentation on Liquid Phase Sintering in Gravity and Microgravity Environments (80NSSC19K0231 ; NNX16AK21G)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	co-investigator Eugene A Olevsky, SDSU, Elisa Torresani, SDSU, MS 1326		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2010 Materials Science NNH10ZTT001N
Start Date:	07/01/2016	**End Date:**	06/30/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 6/30/2021 per NSSC information (Ed., 8/12/20) NOTE: Extended to 6/30/2020 with new grant number 80NSSC19K0231 per NSSC and MSFC personnel (Ed., 2/21/19)		
Key Personnel Changes/Previous PI:	2020 report: Elisa Torresani, Ph.D., will be CoInvestigator in summer 2020.		
COI Name (Institution):	Olevsky, Eugene Ph.D. (San Diego State University)		
Grant/Contract No.:	80NSSC19K0231 ; NNX16AK21G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE (Ed., 2/21/19): Extended to 6/30/2020 with new grant number 80NSSC19K0231 per NSSC and Marshall Space Flight Center (MSFC) personnel. NOTE (Ed., June 2016): Continuation of "Multi-Scale Modeling and Experimentation on Liquid Phase Sintering in Gravity and Microgravity Environments", grant #NNX14AB31G with the same Principal Investigator. A flight research project is in progress investigating the gravity-dependent densification, distortion, microstructure development, and the related macroscopic shape evolution of multi-phase powder alloys during liquid phase sintering (LPS). The research is aimed at achieving two critical goals: (i) in-depth analysis of the LPS induced pore-grain structure evolution by the deconvolution of the gravity impact and (ii) exploring sintering under microgravity conditions as the basis for techniques applied to in-space fabrication and repair. Indeed, future NASA efforts to extend human exploration back to the Moon and beyond will require development of processes that permit particle-based fabrication under reduced gravity conditions. This capability is needed to reduce resource requirements for extraterrestrial construction and in space repair. Many proposed in-space fabrication and repair techniques involve particulate materials processing based on sintering, but limited data are available to assess the viability of these proposed net-shape techniques under reduced gravity. The research team has developed a computer finite element code for the multiple scale simulation of sintering, embracing both microstructure development and macroscopic densification and associated shape distortion. Liquid phase sintering experiments on Earth and in microgravity are being performed using a variety of tungsten heavy alloys (W-Ni-Cu-Mn) to validate the sintering model framework. The anticipated research outcomes will be relevant to current and future space exploration needs for habitat creation, extraterrestrial exploration, and vehicle repair activities during trips to Mars or the Moon.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research is focused on liquid phase sintering, one of the most widely used manufacturing routes for ceramics, cemented carbides, powder metals, and composites. In the USA this is responsible for approximately $25 billion in annual component production. The products are applied to such diverse fields as metal cutting, oil well drilling, automotive engines, electric motors, computer heat sinks, jet engines, medical tools, skeletal implants, exercise monitors, and even wrist watches (especially the newest cell phone coupled devices). These components are designed for high performance and high precision, and are often fabricated in large production quantities. The net-shape fabrication, where final size and shape is achieved without machining, is a challenge in sintering since often dimensions change 15 to 20% due to shrinkage in going from the powder to the dense component. During that high temperature sintering the component strength drops considerably and is subject to gravity effects. Thus, computer models of the sintering process are vital to proper tool and process design. Further, to attain specific mechanical, thermal, electrical, or other properties requires further attention to microstructure control. This research provides a verified computational model to enable prediction of final size and shape for different gravitational fields, using a multiple scale model with hopes to extend from geometry to property predictions. The robust inclusion of gravitational gradients, liquid-solid migration, and substrate friction, the computer simulation will be useful for predicting component attributes over a range of gravitational conditions, as required for eventual additive manufacturing as potentially envisioned for fabrication on the moon, other planets, or in microgravity, for example, in building structure on the lunar surface or repairing space vehicles. Of course, this has benefit on Earth with the evolution of a computational means to predict high tolerance shapes by sintering. We are adding the missing gravitational term to computer simulations, typically finite element analysis tools, for predicting final shape and size.
	This research effort is in the final flight phase with two of seven sample cartridges successfully processed through short and long sintering cycles on the International Space Station to provide the needed microgravity data. The first successful set of sintering samples were returned to the investigation team in early 2020 for direct comparison to parallel ground based sintered samples. The samples constitute a range of tungsten heavy alloy compositions especially formulated for these experiments, consisting of various tungsten, nickel, copper, and manganese contents. The alloys represent differences in the solid-liquid ratio at 1200°C, with further differences in the liquid-solid dihedral angle as adjusted by the nickel-copper ratio. The microgravity runs that are being compared with previously completed ground experiments. Some of the ground sintering experiments were in a laboratory batch furnace or a constant heating rate dilatometer, and some were sintered in a ground version of the flight furnace. This comparison with gravitational conditions provides needed parameters for the modeling sintering in different gravity conditions. Key concerns are proper prediction of densification and distortion. The starting compacts are nominally 69% dense, 12 by 12 mm right circular cylinders. They were hydrogen pre-sintered to provide handling strength. Attention is now moving to quantification of microstructure development, since the tungsten particles dissolve and reprecipitate in the liquid during sintering, leading to substantial grain growth during sintering. The findings will fill in missing data in applying finite element models to predict final size, density, shape, and properties (largely dependent on microstructure). The finite element models, popular for modeling sintering, require a rheological approximation during the sintering cycle, with the effective viscosity falling to approximately 1 GPa·s, a value similar to Silly Putty. The research manipulates the sample and sintering cycle to broadly analyze the pore-liquid-grain structure evolution. Processing in microgravity allows extraction of the gravitational deviatoric term as related to densification. It is expected densification will be more difficult in microgravity sintering since the compressive stress is missing. This has serious implications for difficulty reaching technical goals (mechanical properties for example) using in-space fabrication or repair concepts. Models are needed to allow insightful modifications to the green body and heating cycle to properly afford net-shape fabrication f where reduced gravity conditions exist. Remote fabrication options greatly reduce resource requirements for extraterrestrial construction and repair using particulate materials and sintering. To date, limited data are available to assess the viability of proposed net-shape techniques under reduced gravity. The research team has developed a finite element simulation for sintering, embracing both microstructure development and macroscopic densification. It successfully predicts shape distortion observed on Earth. The ongoing microgravity experiments provide the test data to validate the sintering models in a different gravitational environment. The research outcomes are relevant to space exploration and needs for habitat creation, extraterrestrial exploration, and vehicle repair activities during trips to Mars or the Moon. During the reported period the following research activities have been carried out: • Seven cartridges with 49 samples of different composition have been on the International Space Station for some time awaiting furnace and crew time.

Task Progress:	• Two microgravity experiments with seven samples each have been successfully performed successfully. • One cartridge has returned to Earth and is in analysis by the investigation team, and no damage of significance was noted (other than a few divots on the quartz containment and slight vapor staining on the alumina ceramic crucibles). • Modeling efforts are largely completed and await final densification and distortion measurements from samples fired in the Low Gradient Furnace (LGF) to enable refined constitutive equations. • Comparative reports have analyzed ground samples and microgravity samples for dimensional change, densification, mass change, and final profiles; an early document has captured these findings, but nothing has been offered for publication. • Quantitative microstructure analysis of the flight samples is awaiting completion of the coronavirus quarantine, but ground samples are fully documented for microstructure versus location within the samples, documenting gravitational gradients in density, grain size, pore size, and distortion versus tungsten content and dihedral angle (adjusted via copper-nickel ratio). The effect of gravity was studied during liquid phase sintering considering using W-Ni-Cu-Mn compositions. Two compositions changes are in the tungsten (solid) content and in the nickel to copper ratio. The starting compacts are about 69% dense, a value typical to sintering tungsten heavy alloys. The gravity effects during liquid phase sintering for the tungsten alloys is adjusted for the composition changes. The model is implemented in COMSOL software following the ideas previously published in Acta Materialia 65 (2014):176-184. The constitutive equation for sintering densification includes rheological parameters to model the solid-liquid system at temperature reflective of liquid phase sintering. The model relies on the sintering stress to induce densification. This stress arises from capillary attraction between solid grains, with η and ζ are the shear and bulk viscosities. Both change during sintering with porosity, solid volume fraction, and the temperature T. The sintering stress represents the driving force of the densification and is proportional to the surface energy. It arises from the inherent curved surfaces associated with small particles. Prior models have assumed an average value, but we find the stress varies with the degree of sintering. Sintering is treated as a viscous flow, giving an equation of momentum conservation as expressed in terms of the local density and gravity vector. The model accounts for solid settling within the component, evident under Earth gravity sintering conditions, giving a concentration-induced atomic diffusion that relates to viscosity, which is easier to envision. The gravity induced solid settling (tungsten is almost twice the density of the liquid) is treated using Stokes law; neglecting edge effects and grain aggregation (sinter bonding) that alters the settling velocity of a single grain. From Stokes settling velocity, the flux of solid grains settling depends on the gravity field and effective viscosity. An interesting interplay of forces arises to cause grain settling and liquid displacement, but that is absent in microgravity. Part of the difficulty is the solid phase forms bonds between grains, but solvation on liquid formation dissolves those bonds, leading to a transient from pure solid, crowded suspension, semisolid, and finally skeletal solid with interlaced liquid over about 5 to 10 minutes of sintering time. Additive manufacturing is a technology that allows component fabrication in remote locations. The possibility to fabricate components with complex shape without traditional manufacturing facilities is useful for remote repairs. Components obtained via binder jetting are a focus since the ideas are established and rely on sintering ideas. During sintering the gravitational forces modify the final component shape to induce distortion aligned with the gravity vector. Sintering densification and distortion are expected to follow a novel trajectory in the absence of gravity. For example, ponder efforts to apply additive manufacturing in microgravity. On Earth we find densification occurs first followed by distortion. The conjecture in reduced gravity is the components will not densify (pores coalesce) and due to missing particle contact forces, yet the particles will easily slide to allow distortion. In COMSOL the 3D geometry of the components is implemented and models the experimental results with accuracy. The current microgravity trials will provide critical data to include variable gravity and microgravity in the model. The effect of gravity is examined for two cases: First, is examination of the gravity effects during the liquid phase sintering of samples sintered in the ground-proof LGF experiments. These employ various tungsten alloys with intentional variations in solid content and dihedral angles. Second, is study of the gravity effects during sintering of components obtained using 3D printing technology, including the study of the gravity-induced shape distortions during the debinding/sintering stage.
Bibliography Type:	Description: (Last Updated: 03/27/2025)
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